AZAM MOHAMED

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EDUCATION

University of California, Santa Cruz, Santa Cruz, CA Sep 2023 – Aug 2026
B.S. Computer Engineering, Minors in Electrical Engineering and Computer Science
Relevant Coursework: ECE 118: Mechatronics, Embedded Systems, Computer Architecture, CSE 134: Operating Systems, CSE 130: Computer Systems, Data Structures

WORK EXPERIENCE

UC Santa Cruz jLab (Josephson Lab), *Undergraduate Researcher, ENTS Project*, Santa Cruz, CA Jul 2026 – Present
- Own end-to-end data validity for a deployed ENTS node running ENTS-tock on Tock OS, where a node can appear healthy while emitting silently corrupt samples.
- Verify the telemetry path by decoding the node's nanopb Protocol Buffer packets and confirming each field maps to a plausible sensor value, isolating whether corruption originates in firmware, transport, or the backend.
- Instrument the firmware with a local microSD log to capture intermittent faults at the moment they occur rather than inferring them afterward.

UC Santa Cruz, *Undergraduate Researcher, Independent Study*, Santa Cruz, CA Jun 2026 – Present
- Configuring and testing the department's PICO compute cluster for Kubernetes research workloads under Prof. Abel Souza.
- Evaluating Incus system containers and VMs on Linux to build reproducible, isolated experiment environments, prototyping container and VM provisioning workflows.

Santa Cruz Beach Boardwalk, *Rides Operator*, Santa Cruz, CA Feb 2026 – Jun 2026

PROJECTS

Autonomous Competition Robot ("Sluggers of the Lost Goal" Competition) *Embedded C, ESP-IDF, HSM, Analog Circuits*
- Owned the ESP32-C3 subsystem on a dual-microcontroller robot (ESP32 + PIC32 Uno32), driving two brushless flywheel ESCs with 50 Hz software PWM and defining a single-byte UART protocol that let teammates develop the driving state machine independently against a stable interface.
- Took ownership of the failing navigation HSM near checkoff, diagnosing faults across state logic, sensor polarity, and motor direction (inverted pivot/spin, drifting tape-following, phantom bumper events), turning an unreliable run into a robot that consistently completed the full field sequence.

Bluetooth HID Mouse (ESP32-C3) *Embedded C, ESP-IDF, BLE, I2C, IMU*
- Built a wireless Bluetooth HID mouse on the ESP32-C3 driven by an ICM-42670-P IMU, converting board tilt into real-time cursor motion at a 100 Hz update rate with no perceptible lag on the host.
- Brought up a custom I2C driver on GPIO7/GPIO8 after the default ESP-IDF pin mapping failed to talk to the IMU, and enabled low-noise accelerometer mode to get stable, drift-free tilt readings.
- Reconfigured the ESP-IDF BLE HID stack from its media-role example into a mouse device with a 3-byte report descriptor, so the host recognized and tracked it as a standard pointing device.

Pintos Kernel Scheduler *C, OS Internals, Thread Scheduling, Interrupt Handling*
- Reworked Pintos thread sleeping to use an ordered sleep_list with a custom wakeup_less comparator, replacing busy-waiting with efficient timed wakeups inside the kernel.
- Woke sleeping threads from the 8254 timer interrupt handler by popping due threads from the sorted list front, disabling interrupts around insertion to avoid races with the ISR; passed all six alarm tests under QEMU.

Thread-Safe Binary Search Tree *C, pthreads, Mutexes, Valgrind*
- Implemented a thread-safe binary search tree in C with fine-grained mutex locking for concurrent insert, search, and delete, hardened against races and leaks with Valgrind and multi-threaded stress tests.

2x2 Matrix Solver ASIC (TinyTapeout, SKY130) *Verilog, cocotb, OpenLane, Fixed-Point*
- Designed a sequential 2x2 linear-system solver (Ax=b, Q4.4 fixed point) in Verilog for the TinyTapeout SKY130 flow, fitting the datapath and a 12-state control FSM in a 1x1 tile targeting 50 MHz.

SKILLS

Languages: C, C++, Embedded C, Python, Verilog, Assembly (RISC-V)
Embedded: ESP-IDF, FreeRTOS, Tock OS, RTOS, ESP32, PIC32, MPLAB X, FSM/HSM Design, Real-Time Systems, I2C, SPI, UART, GPIO, PWM, BLE, Protocol Buffers (nanopb)
Systems: Linux, Operating Systems, Kernel Internals, Concurrency, POSIX Threads, Docker, Kubernetes, Incus
Tools: Git, GDB, Valgrind, Oscilloscope, Logic Analyzer, cocotb, Vivado, OpenLane
Activities: Member, UC Santa Cruz High Performance Computing Club